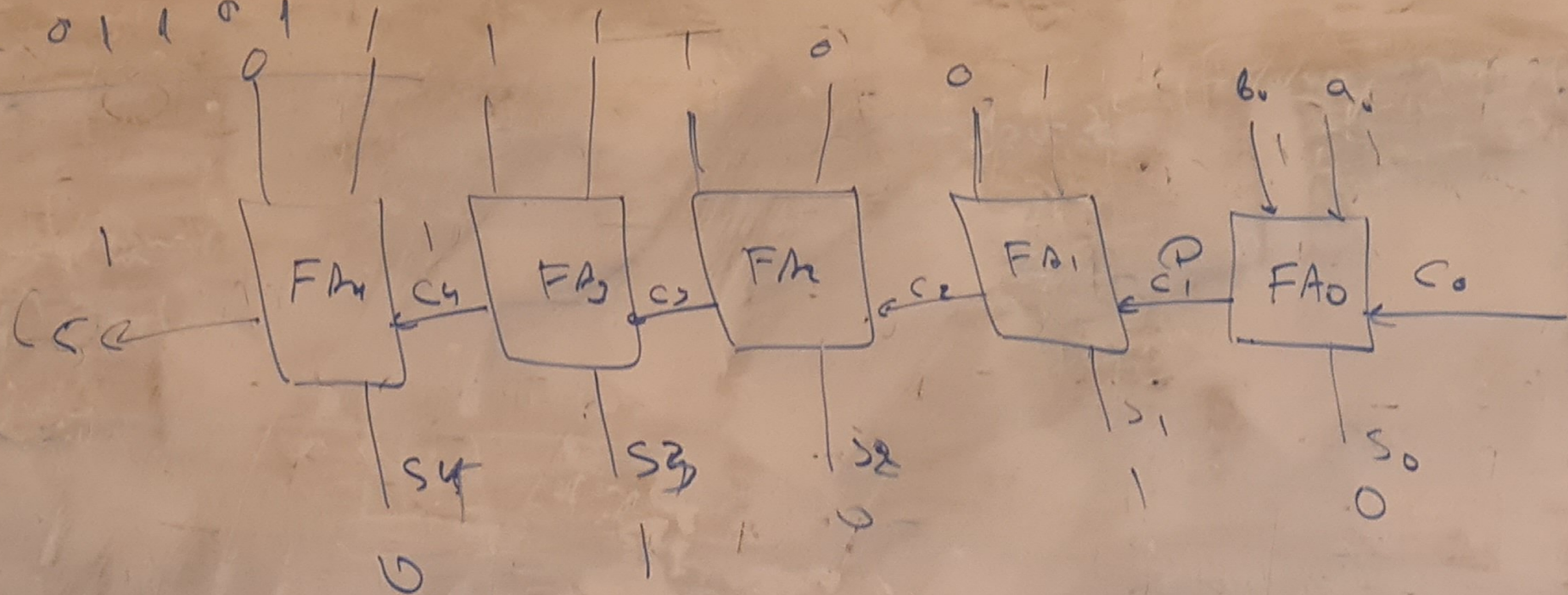


$$A = \begin{matrix} a_4 & a_3 & a_2 & a_1 & a_0 \\ 1 & 1 & 0 & 1 & 1 \end{matrix}$$

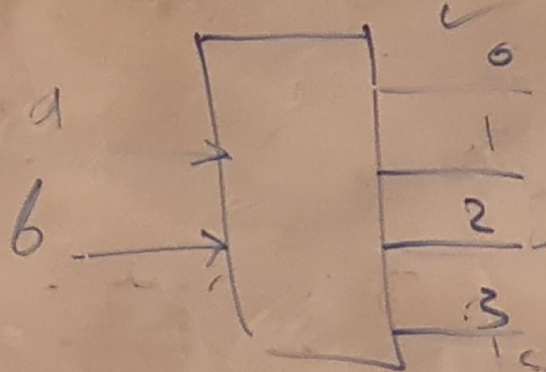
$$B = \begin{matrix} b_4 & b_3 & b_2 & b_1 & b_0 \\ 0 & 1 & 1 & 0 & 1 \end{matrix}$$



$$\begin{array}{r} 10 \\ 1010 \\ \hline \end{array}$$

Decoder

$n \rightarrow 2^n$
Input output

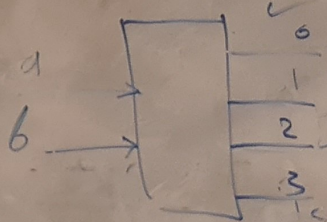


10

		output			
Q	a	D ₀	D ₁	D ₂	D ₃
		0	1	0	1
0	0	1	0	0	0
1	0	0	1	0	0
2	1	0	0	1	0
3	1	0	0	0	1

Decoder

$n \rightarrow 2^n$
Input output



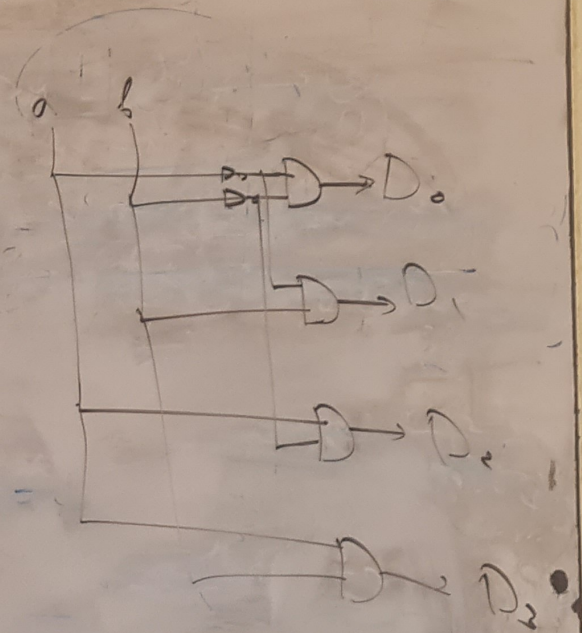
$$D_0 = \bar{a} \bar{b}$$

$$D_1 = \bar{a} b$$

$$D_2 = a \bar{b}$$

$$D_3 = ab$$

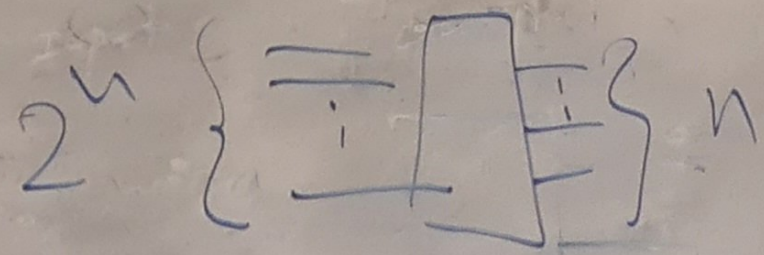
		output			
a	b	D_0	D_1	D_2	D_3
0	0	1	0	0	0
0	1	0	1	0	0
1	0	0	0	1	0
1	1	0	0	0	1



$$= 40.24 \times 10^6 \text{ rad/s}$$

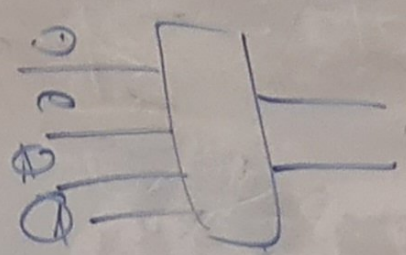
Active and Passive Node

Encoder



10

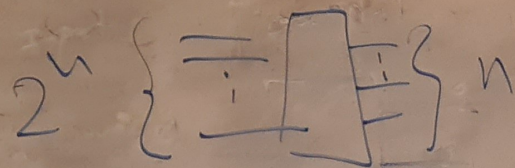
2^n



2^n

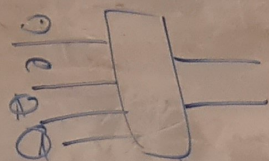
a_3	a_2	a_1	a_0	X	Y
0	0	0	1	0	0
0	0	1	0	0	1
0	1	0	0	1	0
1	0	0	0	1	1

Encoder



10

2^n



2^n

a_3	a_2	a_1	a_0	X	Y
0	0	0	1	0	0
0	0	1	0	0	1
0	1	0	0	1	0
1	0	0	0	1	1

